

Engaging Racing Fans through Offline E-racing Spectator Experience in AR

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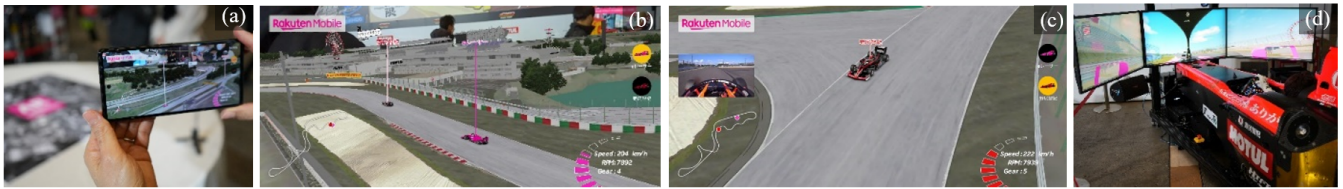


Figure 1: Overview of our AR spectator experience. (a) Visitors watch the race in AR. (b) A screenshot shows the e-racer's game car (pink) racing against the pre-recorded data from a real race car (red/black). (c) Another screenshot shows the on-board camera video appears when the view target is switched to the virtual race car. (d) The e-racer's simulator is placed next to the spectator experience area, and the game car data is sent to the spectator app in real time.

Abstract

In this work, we proposed a new spectator experience for engaging racing fans, even when they cannot be at the race track or, on non-racing days. Audiences can watch an e-racer driving on a simulator race against the pre-recorded car data from an actual race at the Suzuka Circuit. Using mobile phones as AR displays, users can watch the two cars race against each other on a virtual 3D race track, replicating the real race track in AR. This experience was deployed at a motor show, allowing visitors to try this experience. Feedback was also gathered from the visitors, with many of them reporting feeling excited with such kind of racing, and feeling a sense of immersion with such AR experience, indicating strong potential of this new spectator experience.

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1 Introduction

In recent years, interest in enhancing the spectator experience by blurring the lines between spectators and players has surged [Pati-banda, 2023]. This is crucial for eSports and traditional sports events,

where fan engagement drives substantial revenue. Event organizers are increasingly taking on the challenge of creating innovative spectator experiences to attract and engage new and existing fans by employing novel technologies and digital interactions. For instance, new broadcasting systems have been developed to make remote viewing more realistic and immersive [Muto, 2021]. Event organizers are increasingly incorporating augmented reality (AR) to enrich the spectator experience with various digital content. For example, AR can overlay match scenes with player statistics, providing spectators with deeper insights into the game [immersive.io, 2020].

In this work, we leverage AR technology to enhance the spectator experience for an eSports motorsports combination. We not only display statistics about the cars but also introduce interactive features such as adjustable viewing angles and zoom capabilities to immerse fans further into the game. This spectator experience is built upon our previous work [Wu et al., 2023], a "Mixed Reality Racing" experience that merged real and virtual racing, allowing professional drivers and e-racers to compete in real time. With a new form of racing experience, a unique spectator experience was also introduced. The virtual game car was brought out into the real world using AR, enabling spectators to watch the virtual game car compete with the real race car. While our previous work focused on real-time spectator experiences during the race event, our current work aims to create an engaging experience on non-race days and for off-site events or promotions.

Our updated AR spectating system uses recorded data from a real race car, and real-time data from a racing simulator (which is the basis for the virtual game car). This allows e-racers to compete with the real race car at any time. In addition, instead of using Visual Positioning System (VPS) for placing AR content in the real world at the race track, we shifted to using a simple AR marker, for

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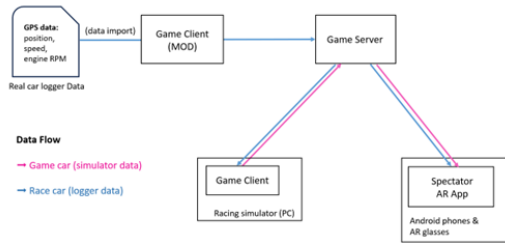


Figure 2: The data flow of our AR spectator app.

placing a virtual 3D Suzuka Circuit, making it easier to view the race at a pre-defined location such as a table.

2 SYSTEM VIEW

The system utilizes car logger data to bring the real SUPER FORMULA race car, which was competing at an actual race to our virtual AR race track. The recreated “race car” can be seen on both the race simulator and the AR spectator app. Conversely, the “game car”, controlled by an e-race driver in the race simulator, can also be seen in our AR app. This system enables e-racers to compete and audience to spectate the race through our AR app. The racing game used in the simulator was “Assetto Corsa”, which includes a digital twin version of the Suzuka Circuit, the real race car, and the game car. The spectator app was developed with Unity and deployed on Android mobile phones. Digital twins of the Suzuka Circuit, the race car, and the game car were all imported into the Unity app. The logger data, stored in a JSON file, encompasses various sensor data from the real race car, including position, speed, and engine RPM, etc., is imported into a client extension or “MOD”, which converts the real race car data into virtual race car data prior to transmitting it to the game server. In Figure 2, the real car data is sent to the game client and the AR spectator app (blue lines). This data is then used to animate a 3D car model, thereby recreating the race car as AR in the spectator app, and as an opponent for e-racers within the game. On the other hand, the game car, controlled by an e-race driver, is sent via the game server to the AR app (pink lines). This allows spectators to view both the virtual race car and the virtual game car competing on a 3D race track through AR devices.

Furthermore, telemetry data from both the race car and the game car were displayed on a Head-Up Display (HUD) to enhance the viewer’s experience and provide insightful information within the AR spectator app. In addition, viewers can toggle to change the focus on the different cars (upper right corner of figure 1.b.). Lastly, on-board camera footage (figure 1.c.) from the real race event was also displayed on the HUD, giving users a more immersive experience and a sense of being in the driver’s seat of the recreated race car.

3 USER EXPERIENCE QUESTIONNAIRE

Our AR spectating system was introduced at the TOKYO AUTO SALON 2023, a leading global motor show. A total of 69 visitors to the event who tried our system responded to our questionnaire, and we highlight major findings in Figure 3.

Q1 assessed the hedonic qualities of this new spectating experience, and the result is very positive, with 84% of respondents agreeing (including both “strongly agree” and “agree” responses) with the question. Comparing Q2 and Q3, 7% of respondents who initially answered “neutral” shifted to “agree” and “strongly agree”,

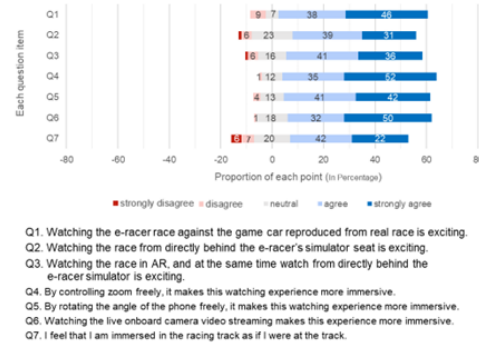


Figure 3: The questions for evaluating our AR spectator app.

indicating that combining AR viewing with race simulator viewing could enhance the overall spectator experience compared to merely viewing the race simulator alone. This suggests the potential for integrating our primary AR spectator app with other experiences to improve overall user engagement. Q4, Q5 and Q6 examined specific elements of our AR app to determine their impact on the immersive experience. Q4 and Q5 focused on viewpoint control, with both free zoom in/out and viewing angle change receiving positive feedback (Q4: 87%, Q5: 83% of respondents agreed). Q6 addressed the contribution of the on-board camera video, with 82% of respondents agreeing that the video playback made the experience more immersive. This likely serves as a useful hint for users to understand the game car’s performance against the race car.

Finally, Q7 examined the overall feeling of immersion. The result was positive, with 64% of respondents agreeing with the question. However, this was noticeably lower compared to the specific features, indicating that other factors may need to be considered. Based on respondents’ feedback, visual quality, choice of AR display, and certain camera angles, such as the driver’s view which could not be adjusted by moving AR devices, have the potential for improvement to increase the immersiveness of the spectator app.

4 CONCLUSION AND FUTURE WORK

We proposed an AR spectator experience that allows users to watch an e-racer compete with a race car in AR, providing offsite fans the opportunity to enjoy races even on non-race days. We investigated how this AR experience enhances excitement, how it changes the traditional spectator experience, and discussed the contributions of each app feature. Insights revealed that controlling viewing angles and integrating various viewing experiences enhanced interactivity and excitement. Visitors’ feedback suggested that improving visual quality, increasing the variety of viewing angles, and offering choices in AR displays could further enhance immersion. These findings guide us in refining our system to provide more immersive and engaging content for fans even on non-race days.

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